

DONGU KIM

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RESEARCH INTERESTS

Storage disaggregation, hardware/software co-design, kernel-bypass storage and networking stacks, RDMA virtualization, in-network computing, and LLM serving

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) 2022 - 2024
M.S. in Computer Science and Engineering

Inha University 2015 - 2022
B.S. in Electronic Engineering

EXPERIENCES

Mangoboost Inc. Apr 2024 - Present
System Software Engineer

· **NVMe-over-TCP Initiator (NTI): Hardware-offloaded Virtualized NVMe/TCP Accelerator** (*flagship product; published at ISCA '26 Industry Track*)

- Designed and implemented a host-platform-agnostic NVMe/TCP Initiator (NTI) software stack, with an ARM-FPGA-accelerated implementation that offloads TCP and NVMe/TCP processing to hardware while supporting zero-copy data transfer, fault tolerance, and resilience
 - ✓ Implemented full NVMe/PCIe device emulation toward the host NVMe driver, enabling host-platform-agnostic compatibility across Windows, Linux, vhost, and virtualized environments; certified by UNH-IOL
 - ✓ Designed and implemented system-level robustness mechanisms to tolerate datacenter failure scenarios, including network faults and power shutdowns

· **MASS: Scalable, High-Performance Storage System for AI**

- Design and implement an RDMA-based high-performance storage cluster using DPU-JBoFs and flash arrays to store and offload LLM context data, such as KV caches, enabling efficient sharing across GPU clusters
 - ✓ Design and implement a Kubernetes-based storage resource allocation system built on top of Distributed Asynchronous Object Storage(DAOS) to manage storage resources across distributed clusters
 - ✓ Integrate FPGA-based in-house RDMA and DPU cards into the storage cluster and conduct performance evaluations across a variety of workloads
 - ✓ Design and implement a lightweight distributed storage system based on NVMe-over-Fabrics (NVMe-oF), integrating it with LLM serving frameworks such as Mooncake and LMCache

Next Generation Networks and Systems Lab (*Advisor: Youngbin Im*) Mar 2022 - Feb 2024
Research Assistant

· **FlexServer: High-performance Content Servers Accelerated by DPU (400 Gbps Throughput)**

- Designed a DPU-based multi-tier content caching system that reduces data copying in host TCP stacks and accelerates content delivery through an asynchronous I/O stack, achieving up to 351 Gbps, 94% of 400 GbE line rate, and 4.4× higher throughput than the state-of-the-art IO-TCP baseline
 - ✓ Designed and implemented a SmartNIC-based software stack on Netronome Agilio and NVIDIA BlueField platforms, leveraging P4, microC, DPDK, and DOCA for packet control

· **Core Principles of Cloud Computing**

- Studied cloud infrastructure fundamentals, focusing on resource virtualization, SDN (Software Defined Networking), and storage orchestration.

PUBLICATIONS

1. Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, Youngwoo Jeong, **Dongu Kim**, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim, “BoostXTM-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration,” *ISCA '26 (Industry Track)*, 2026.
2. **Dongu Kim**, Seyeong Park, Seyeon Kim, Sangtae Ha, Youngbin Im, “FlexServer: A High-Performance Content Server Leveraging Asynchronous I/O and SmartNICs,” *Preprint*, 2024.

PATENTS

1. Youngbin Im and Dong U Kim, “METHOD AND APPARATUS FOR IMPROVING KEY VALUE STORE PERFORMANCE THROUGH OFFLOADING OF SMART NETWORK INTERFACE CARD,” KR Patent Registered No. 10-2724-6120000, Oct. 2024

TECHNICAL SKILLS

- **Languages:** C/C++, Go, Python, P4
- **Systems & Frameworks:** Linux, DPDK, SPDK, Kubernetes, DAOS, Mooncake
- **Core Technologies:** RDMA, TCP, PCIe, NVMe, NVMe-oF
- **Hardware Platforms:** NVIDIA BlueField, Netronome Agilio SmartNIC, Xilinx FPGA

EXTRACURRICULAR ACTIVITIES

Competitions & Projects

· **The World Embedded Software Contest 2019**

Apr 2019 - Dec 2019

- Developed multithreaded software for an autonomous RC car featuring real-time obstacle avoidance; gained in-depth expertise in concurrent programming and thread synchronization
- Integrated various sensors—including LiDAR and Ultrasonic—with the software architecture to improve the accuracy of environmental mapping

Volunteer Work

· **Freewheel**

Feb 2020 - Jan 2021

- Served as a volunteer mentor, sharing specialized knowledge about my major to help middle and high school students explore potential career paths and gain a deeper understanding of the field